

1. The switching waveforms of one power semiconductor device are shown in Fig.1. If $V_o=700V$, $I_o=40A$, voltage drop of the switch $V_{ce(sat)}=2V$. The turn-on delay and turn off delay time are neglected. Please answer the following questions: (7 points)

某一电力电子器件的开关波形如图 1 所示。如果 $V_o=700V$, $I_o=40A$, 通态压降 $V_{ce(sat)}$ 为 $2V$ 。忽略开通延迟和关断延迟时间。回答以下问题。(7 分)

- (1) What is the switching power loss of the switch? (2 points)

开关管的开关损耗是多少? (2 分)

- (2) What is the total power loss of the switch? (2 points)

开关管的总损耗是多少? (2 分)

- (3) Which kind of device is used in this case? Draw the symbol of the device. (2 points)

请问这种情况下使用了哪种器件? 画出对应的符号。(2 分)

- (4) Please write down at least one method to reduce the switching power losses. (1 points)

请至少说出一种减小开关损耗的方法。(1 分)

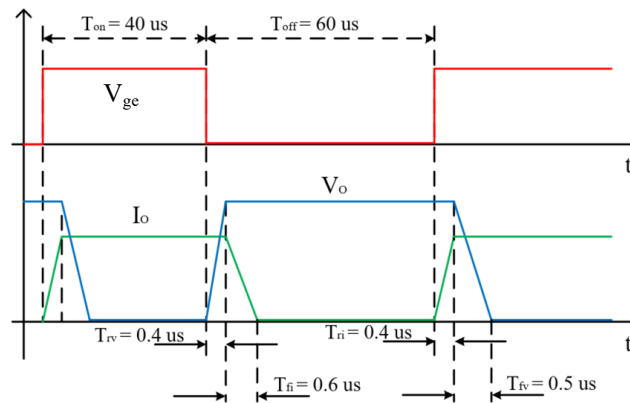


Fig.1 Switching waveforms of one power semiconductor device

答案:

$$(1) P_s = \frac{V_o I_o f_s (T_{rv} + T_{fi})}{2} + \frac{V_o I_o f_s (T_{ri} + T_{fv})}{2}$$

$$f_s = \frac{1}{T_s} = 10 \text{ kHz}$$

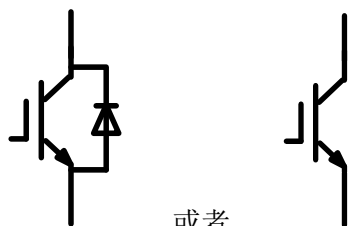
$$P_s = \frac{700 \times 40 \times 10^4 \times (0.4 \times 10^{-6} + 0.6 \times 10^{-6})}{2} + \frac{700 \times 40 \times 10^4 \times (0.4 \times 10^{-6} + 0.5 \times 10^{-6})}{2} = 266W$$

(2 分)

$$(2) P_{cond} = V_{ce(sat)} I_o (T_{on} - T_{ri} - T_{fv}) f_s = 31.28W$$

$$P = P_s + P_{cond} = 266 + 31.28 = 297.28W \quad (2 \text{ 分})$$

(3) 器件类型: IGBT,



符号:

或者

(2 分, 器件类型和符号各 1 分)

(4) 采用开关损耗更小的新器件(GaN、SiC)或通过控制实现软开关, 加缓冲电路等等。 (答对任意 1 个得 1 分)

2. Fig.2 shows Buck converter circuit, and there is a spike in the voltage waveform v_{sw} , as shown in Fig. 3. Please answer the following questions. (9 points)

图 2 是 Buck 变换器电路, 开关管 S_w 的电压波形上有一个电压尖峰, 如图 3 所示, 请回答下列问题。(9 分)

(1) Which circuit can be added to reduce the voltage spike in Fig. 3? Draw this additional circuit in Buck dc-dc converter. (4 points)

请问增加什么样电路可以降低图 3 中的电压尖峰? 在 Buck 电路中画出该附加电路 (4 分)

(2) Explain the function of each component in this added circuit. (3 points)

解释该附加电路中各元件的功能。(3 分)

(3) What function does gate drive circuit have in Fig.2? (2 points)

图 2 中的栅极驱动电路有什么功能? (2 分)

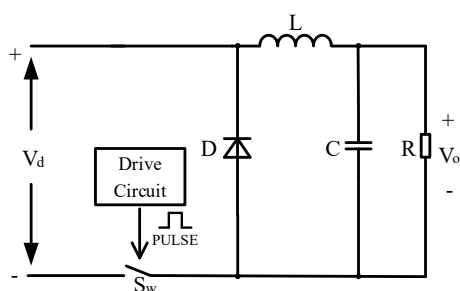


Fig.2

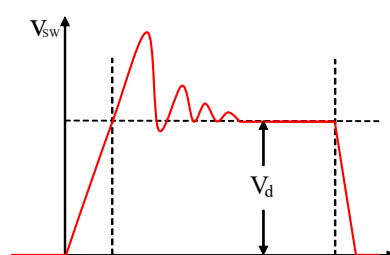
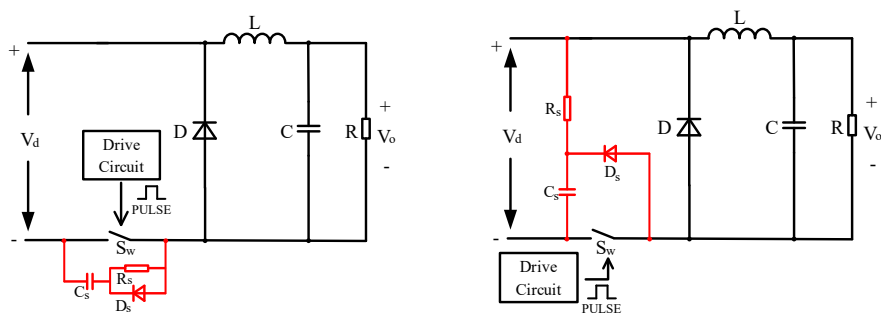


Fig.3

答案:

(1) RCD 缓冲吸收电路(1 分)



或

(电路图 3 分，画出一个即可)

- (2) 在 RCD 缓冲吸收电路中, R_s 提供 C_s 放电通路, 消耗 C_s 上的能量 (1 分); C_s 用于储能, 吸收过电压、降低 dv/dt (1 分); D_s 提供对 C_s 快速充电的通道 (1 分);
- (3) 提供控制电路与主电路的接口, 信号放大, 电气隔离, 驱动信号整形, 过压过流保护等功能。 (2 分, 答对任意 2 个即可)

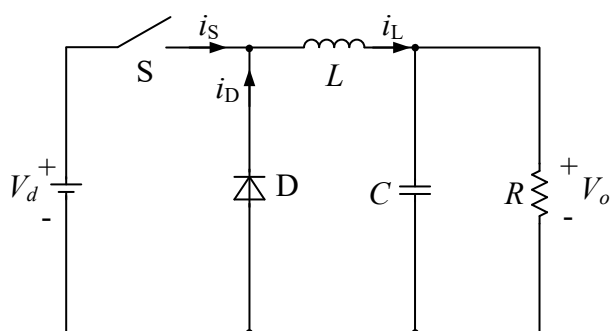
2. (16 points) In a Buck converter, consider all components to be ideal. Assume

$V_d=12.5V$, $V_o=5V$, $I_o=2A$, $f_s=20kHz$, $L=100\mu H$ and $C=470\mu F$:

- (1) Draw the circuit diagram of the Buck converter. (2 points)
- (2) Calculate Δi_L and ΔV_o . (6 points)
- (3) Determine the type of switch S and diode D ? (2 points)
- (4) Draw the waveform of v_L , i_L , i_s , and mark the value. (6 points)

1.

(1)



(2 分)

(2) 由已知得: $V_o = 5V$, $f_s = 20kHz$, $L = 0.0001H$, $C = 470\mu F$,

$$V_d = 12.5V, \quad I_o = 2A, \quad D = \frac{5}{12.5} = 0.4$$

$$\text{有 } I_{oB} = \frac{D}{2f_s L} (V_d - V_o) = 0.9375A \quad (1 \text{ 分})$$

$\therefore$ 电感工作在连续状态 (1 分), 有:

$$\therefore \Delta V_o = \frac{T_s^2 (1-D) V_o}{8LC} = \frac{[1 - (5/12.5)] \times 5}{(20000)^2 \times 8 \times 0.0001 \times 470 \times 10^{-6}} \approx 16.6mV \quad (2 \text{ 分})$$

$$D = \frac{5}{12.5} = 0.4, \quad I_o = 2A$$

$$V_L = L \left(\frac{di_L}{dt} \right)$$

$$\text{当 } t_{on} \text{ 时, } \frac{di_L}{dt} = \frac{12.5 - 5}{0.0001} = 75000A/s$$

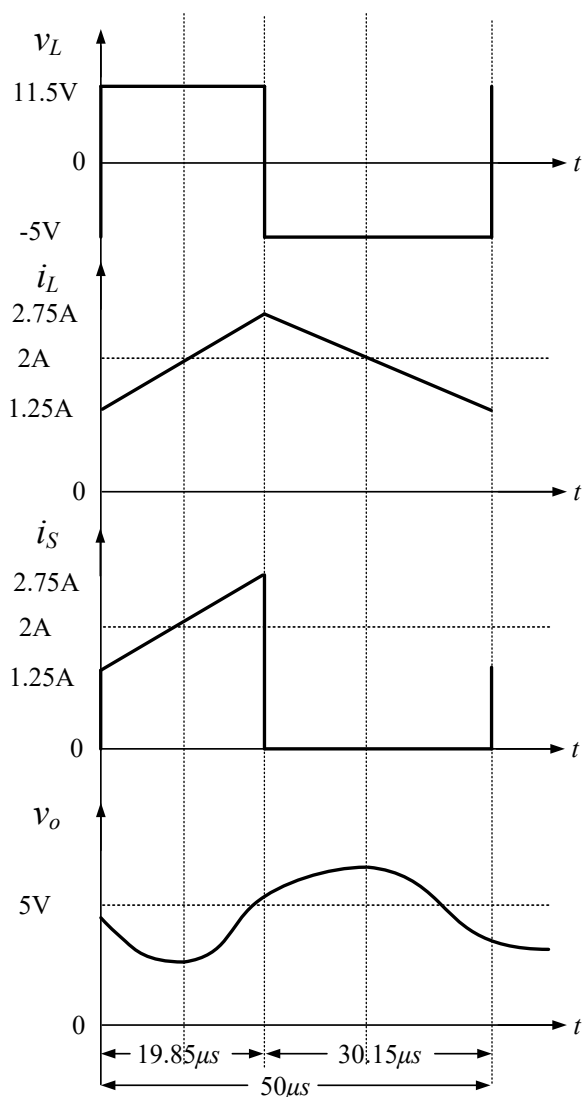
$$\text{当 } t_{off} \text{ 时, } \frac{di_L}{dt} = \frac{-5}{0.0001} = -50000A/s$$

$$D = \frac{5}{12.5} = 0.4, \text{ 所以电感电流的纹波值为:}$$

$$\Delta i_L = D \times T_s \times \frac{di_L}{dt} = 0.4 \times (50\mu s) \times (75000A/s) = 1.5A \quad (2 \text{ 分})$$

(3)

(8 分, 各 2 分)



(4) S 选择 MOSFET (1 分), D 选择肖特基二极管 (1 分), 因为开关频率高达 20kHz, 二极管 D 所承受的电压应力大约为 30V (2 分)

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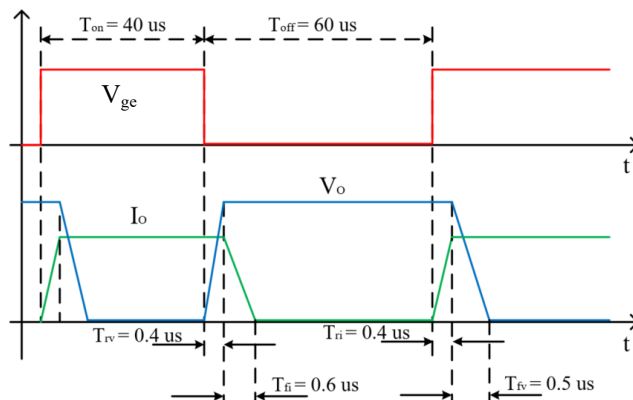


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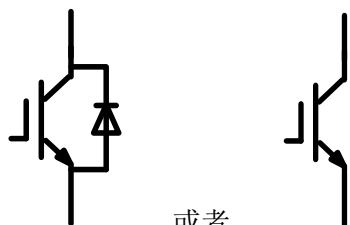
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